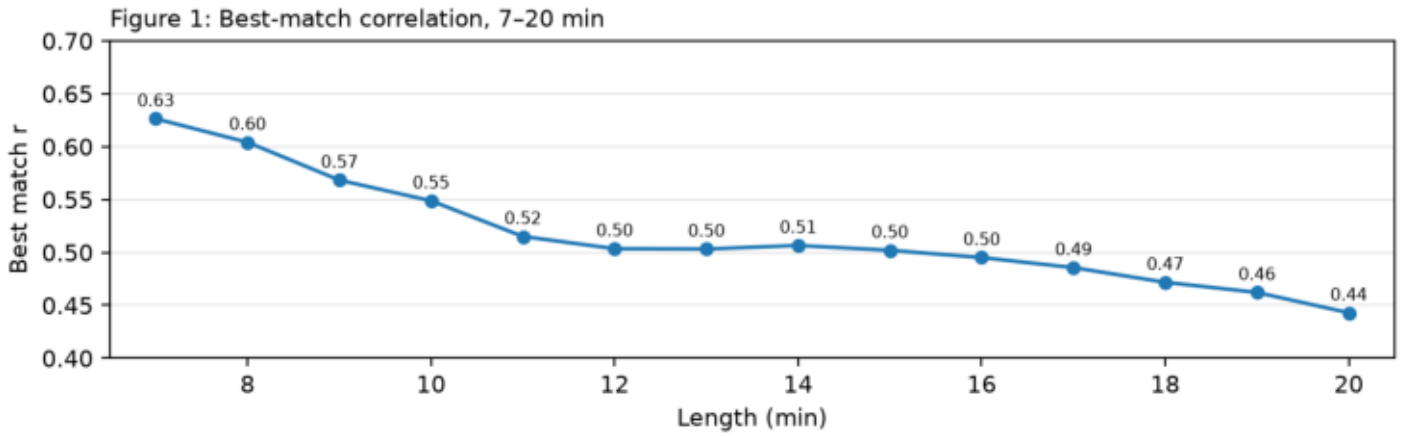
Song structure in a 10-hour humpback recording (LeiData 193500)

Summary

• Song cycle of about 12.8 minutes (fine-scan median at 9–10 min).
• Stable across motif lengths 6–10 min; coarse 2/4/8/16 min do not match.
• Seven 1-minute phrase groups with fixed relative places.
• 20 of 41 songs follow G5 → G2 → G1 → G6 → G4 → G3 → G7 (shuffle p = 0.01).
• Energy minima fall inside the 7-min marker, near the start of the phrase order.

Data and method

File LeiData/SOCAL_H_76_230426_193500_df100.x.x.wav, starting 2023-04-26 19:35:00.
Audio 10 h, mono, 2 kHz. Features: MFCC coefficient 2 at 10 frames/s (360000 frames).
1. Matrix profiles via mpx_fast / mpxForPan with non-overlapping matches; $r = 1 - mp^2/(2w)$.
2. One motif group per length: best-pair seed ($r \geq 0.2$); copies overlap ≤ 10%; threshold lowered from 0.2 until no stretch > 20 min lacks a copy.
3. Neighbouring lengths compared by occurrence counts, KS D, and Wasserstein W.



Length (min)	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Best r	0.63	0.60	0.57	0.55	0.52	0.50	0.50	0.51	0.50	0.50	0.49	0.47	0.46	0.44

Motif length scan

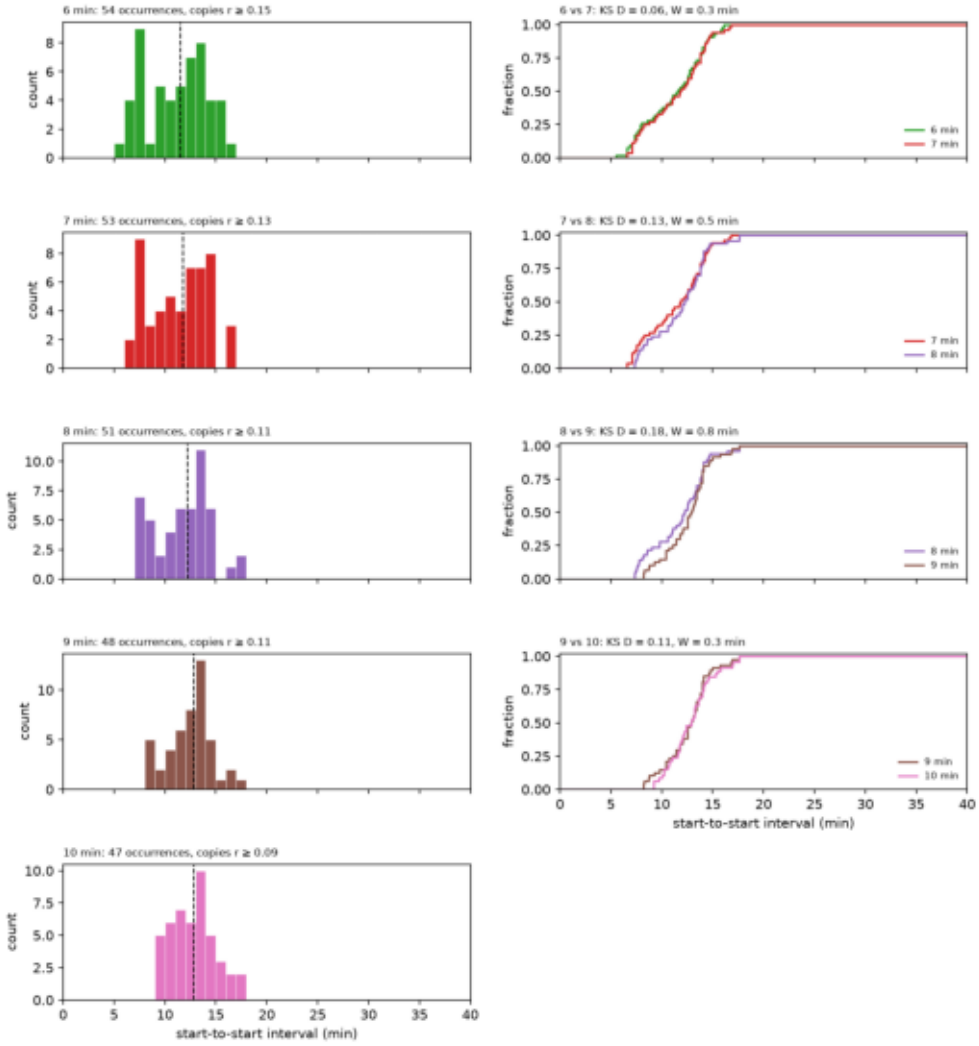
The aim is two neighbouring lengths with the same number of occurrences and similar start-to-start histograms.

Coarse scan: 2, 4, 8 and 16 minutes

Pair	Occurrences	KS D	W
2 vs 4 min	125 vs 66	0.62	4.5 min
4 vs 8 min	66 vs 51	0.36	2.6 min
8 vs 16 min	51 vs 31	0.90	7.5 min

- Short motifs repeat within a song. 2- and 4-minute medians 3.8 and 8.5 min.
- 16 minutes (threshold -1.00) is longer than a song.
- Fine scan: 6–10 minutes. Every neighbouring pair matches.

Figure 2: Start-to-start intervals, 6–10 min



Length	Occurrences	Copy threshold	Median interval (IQR)	Intervals < 10 min
6 min	54	0.15	11.5 (8.0–13.7)	20 of 53
7 min	53	0.13	11.8 (8.5–13.7)	18 of 52
8 min	51	0.11	12.2 (9.7–13.8)	14 of 50
9 min	48	0.11	12.8 (11.1–13.9)	7 of 47
10 min	47	0.09	12.8 (11.4–14.0)	5 of 46

“Same occurrences” counts shorter-length midpoints within 1 minute of a longer midpoint.

Pair	Count difference	KS D	W	Same occurrences
6 vs 7	2%	0.06	0.3 min	51 of 54
7 vs 8	4%	0.13	0.5 min	47 of 53
8 vs 9	6%	0.18	0.8 min	44 of 51
9 vs 10	2%	0.11	0.3 min	44 of 48

Inside the song: 1-minute phrases

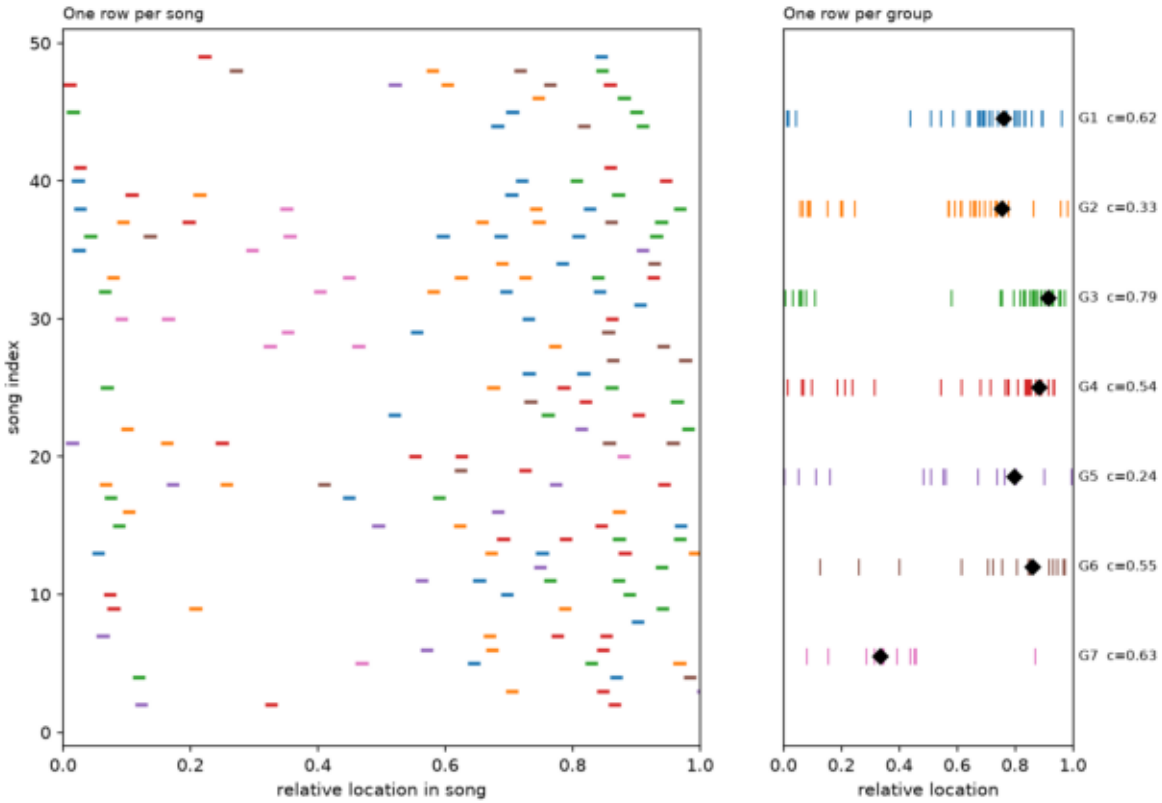
Definitions

• Songs bounded halfway between consecutive markers → 51 songs, median length 12.0 min (range 7.1–15.7).
• Relative location of midpoint t in song k is (t − B_k)/(B_{k+1} − B_k).

Phrase groups

• Seeds and copies need r ≥ 0.35, at most 30 copies per group, no overlaps.
• Found 30 groups; 7 with ≥ 8 occurrences kept as G1–G7.
• Kept groups cover 2.7 h; all groups 3.8 h.

Figure 3: Relative location of the 1-minute phrases



Group	Occurrences	Songs	Seed r	Circular mean	c
G1	30	23	0.78	0.76	0.62
G2	30	22	0.61	0.76	0.33
G3	30	24	0.71	0.91	0.79
G4	30	24	0.66	0.88	0.54
G5	14	13	0.49	0.80	0.24
G6	17	14	0.54	0.86	0.55
G7	12	10	0.50	0.33	0.63

Order of the phrases

Song start moved to the largest empty stretch of relative locations (0.37).

Pairwise order

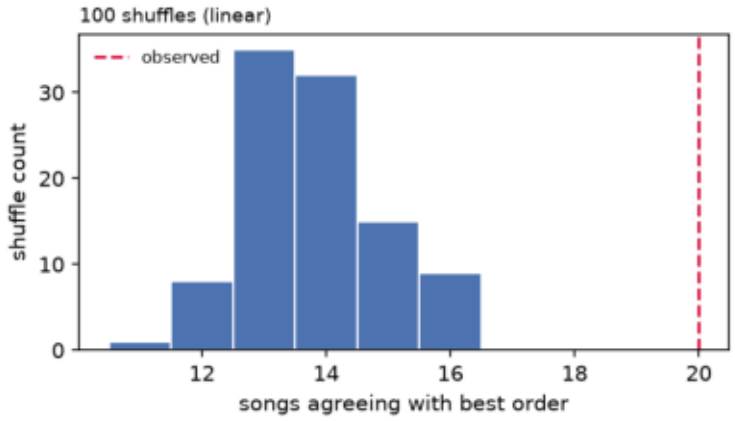
Earlier → later	In this order	p
G6 → G2	2 of 7	1.0000
G1 → G4	7 of 8	0.0703
G6 → G1	0 of 5	1.0000
G6 → G3	5 of 6	0.2188
G1 → G2	2 of 7	1.0000

Consensus order

Figure 4: Consensus linear order and shuffle test

```
Linear best order:
G5 → G2 → G1 → G6 → G4 → G3 → G7

Songs that agree: 20 of 41
Fixed random order mean: 7.6
Shuffled best: mean 13.8, max 16
p = 0.01
```

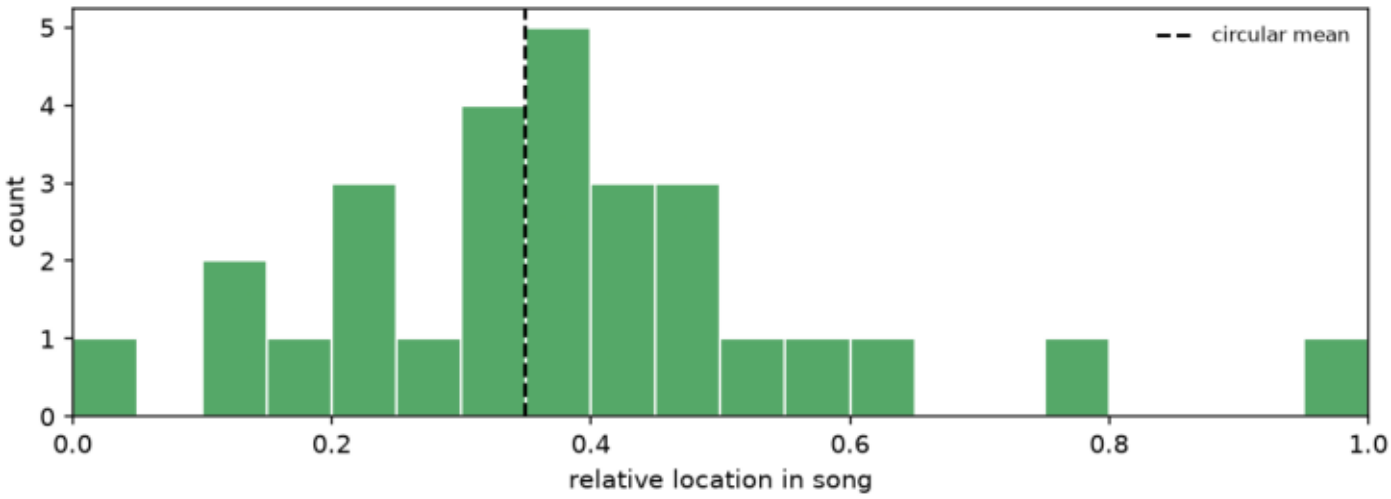


Linear: G5 → G2 → G1 → G6 → G4 → G3 → G7; 20 of 41 agree; shuffled mean 13.8, max 16; p = 0.01.
Circular: G1 → G3 → G2 → G6 → G4 → G7 → G5; 24 of 41; shuffled mean 23.3, max 27; p = 0.40.

Energy minima and the song boundary

Energy: MFCC coefficient 0. Slow curve: subtract 30-min running mean, Gaussian smooth with σ = 4 min. Minima: prominence ≥ 0.01, ≥ 8 min apart.

Figure 5: Energy minima inside the song



Minima 28, of which 28 lie inside a song
Interval between minima median 15.7 min (IQR 13.3–24.2)
Songs with a minimum 28 of 51
Relative location median 0.38 (IQR 0.28–0.46), circular mean 0.35
Concentration R = 0.58, Rayleigh p = 3.3e-05
Inside a 7-min marker occurrence 24 of 28
Offset from nearest marker centre median -1.9 min (IQR -2.5 to +0.2); 20 of 28 before the centre

Caveats and reproduce

Caveats

- Weak copies: thresholds 0.09–0.15 capture every song.
- Minimum interval ≥ 0.9 × motif length.
- Matching uses one MFCC coefficient of audio below 1 kHz.
- Half-song markers distort boundaries and relative locations.
- Greedy phrase groups: G1–G4 can hit the 30-copy cap.

Reproduce (Python)

```
python -m analysis.matrix_profile.replicate_report.report_193500
python -m analysis.matrix_profile.replicate_report.write_experiment_193500_report
```

Profiles and MFCC coefficients are cached under cache_193500/. Kernel: analysis.matrix_profile.mpxForPan / mpx_fast.